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Energy identity for the incompressible Cahn–Hilliard/Navier–Stokes system with non–degenerate mobility

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Abstract. We consider the Cahn–Hilliard/Navier–Stokes system with non–degenerate mobility in the space–periodic case, describing the flow of two viscous immiscible and incompressible Newtonian fluids with matched densities. We identify sufficient conditions on the velocity field for weak solutions to satisfy an energy identity, improving previous results on the literature.

Mathematics Subject Classification. 35Q35.

Keywords. Cahn–Hilliard/Navier–Stokes, Weak solutions, Energy identity, Anomalous dissipation.

1. Introduction

Diffuse interface models have proven effective in explaining the evolution of two or more immiscible fluids. A key strength of these models is their capability to naturally depict topological changes, such as the merging or the separation of droplets. The foundational diffuse interface model in the case of two incompressible, viscous Newtonian fluids, is referred to as Model H (cf. [8]); it was derived also in [7] in the framework of continuum mechanics and it was shown that it is consistent with the second law of thermodynamics. This model gives rise to the Navier–Stokes/Cahn–Hilliard system (see also [3,9]):

$$\partial_t u + \operatorname{div}(u \otimes u) - \operatorname{div}(\nu(c)\mathbb{D}u) + \nabla p = -\gamma \operatorname{div}(\nabla c \otimes \nabla c), \tag{1.1}$$

$$\operatorname{div} u = 0, \tag{1.2}$$

$$\partial_t c + \operatorname{div}(cu) = \operatorname{div}(m(c)\nabla \mu), \tag{1.3}$$

$$\mu = f(c) - \gamma \Delta c, \tag{1.4}$$

where u represents the volume–averaged velocity, $\mathbb{D}u = \frac{1}{2}(\nabla u + \nabla u^\top)$ stands for the viscous stress tensor, p denotes the pressure, and $c \in [-1, 1]$ is an order parameter linked to fluid concentration (e.g. concentration difference). Additionally, $\nu(c) > 0$ is the viscosity of the mixture, $\gamma > 0$ is a parameter related to the thickness of the interfacial region, $f = F'$, where F represents a homogeneous free energy density, μ is the chemical potential, and $m = m(c) \geq 0$ the mobility coefficient.

One of the fundamental assumptions of this model is that the densities of both components are the same, which restricts the applicability of the model to situations when density differences are negligible; for extensions of the above model in the case of different densities we refer to [2,3,10].

To avoid complications with boundary terms, we restrict ourselves to the case of the 3–dimensional torus $\mathbb{T}^3 := (\mathbb{R}/2\pi\mathbb{Z})^3$. Then, we consider equations (1.1)–(1.4) on $[0, T] \times \mathbb{T}^3$, where $T > 0$, with the initial conditions

$$u(0, x) = u_0(x), \quad c(0, x) = c_0(x) \quad \text{on } \mathbb{T}^3. \tag{1.5}$$

We place the following assumptions:

(A1) $F \in C([-1,1]) \cap C^2((-1,1))$, with

$$\lim_{s\to -1} f(s) = -\infty, \quad \lim_{s\to 1} f(s) = +\infty, \quad f'(s) \geq -\alpha, \text{ for some } \alpha > 0,$$

(A2) ν is a positive constant,

(A3) m is a positive constant.

We note that the most interesting cases in systems of Cahn–Hilliard type concern degenerate mobilities, something that our assumption (A3) excludes and we refer to [2] for a treatment of the incompressible Cahn–Hilliard/Navier–Stokes system with degenerate mobility. Nevertheless, the regularity provided by the existence theory is not enough and the fact that the mobility does not vanish is vital in our analysis.

Denoting by L^2_σ the space of squared integrable solenoidal functions, we define a weak solution of (1.1)–(1.5) as follows:

Definition 1. Let $u_0 \in L^2_\sigma(\mathbb{T}^3;\mathbb{R}^3)$ and $c_0 \in H^1(\mathbb{T}^3)$, with $|c_0| \leq 1$ almost everywhere on $\mathbb{T}^3$. Let assumptions (A1)–(A3) hold. A triplet (u, c, μ) with the regularity

$$u \in C([0,T]; L^2_\sigma(\mathbb{T}^3;\mathbb{R}^3)) \cap L^2(0,T; H^1(\mathbb{T}^3;\mathbb{R}^3)) \tag{1.6}$$

$$c \in C([0,T]; W^{1,q}(\mathbb{T}^3)), \text{ for some } q > 3, \quad \Delta c \in L^2(0,T; L^6(\mathbb{T}^3)) \tag{1.7}$$

$$f(c) \in L^2(0,T; L^6(\mathbb{T}^3)), \quad |c| \leq 1 \text{ a.e. on } \mathbb{T}^3 \times (0,T) \tag{1.8}$$

$$\nabla\mu \in L^2(0,T; L^2(\mathbb{T}^3;\mathbb{R}^3)) \tag{1.9}$$

is called a weak solution of (1.1)–(1.5), if it satisfies (1.1)–(1.4) in $D^*((0,T)\times\mathbb{T}^3)$ and (1.5) in $D^*(\mathbb{T}^3)$ and the energy inequality

$$E(t) + \int_0^t \int_{\mathbb{T}^3} \left(\nu|\mathbb{D}u|^2 + m|\nabla\mu|^2\right) \mathrm{d}x\mathrm{d}t \leq E(0) \tag{1.10}$$

holds almost everywhere in $(0,T)$, where the energy of the system is given by

$$E(t) = \int_{\mathbb{T}^3} \left(\frac{1}{2}|u|^2 + \frac{\gamma}{2}|\nabla c|^2 + F(c)\right) \mathrm{d}x.$$

Weak solutions according to Definition 1 assume a non–degenerate mobility and their existence has been established in [1, Theorem 1]. In the case of a degenerate mobility coefficient, Definition 1 no longer applies and we refer to [2] for an appropriate definition.

Our interest is in the energy study of system (1.1)–(1.4) under assumptions (A1)–(A3) and in particular, the goal is to identify conditions under which the energy inequality (1.10) is satisfied as an identity. The 2–dimensional case has been treated in [1] and comes for free under no additional hypotheses. This is not the case for the 3–dimensional case. In [9], the authors show that in the case of a constant viscosity $\nu(c) = \nu$ and a constant mobility coefficient $m(c) = m$, weak solutions satisfy (1.10) as an equality, under the additional assumptions

$$f' \in C([-1,1]) \tag{1.11}$$

and

$$\nabla u \in L^p(0,T; L^s(\mathbb{T}^3)), \tag{1.12}$$

for specific exponents p and s (cf. [9, Theorem 1.1]).

In the present work, we show that one can relax (1.12) and ask that the velocity field u be only Hölder continuous in space. This idea originated from [5] on the energy study for the Navier–Stokes equations. In particular, we prove the following theorem:

Theorem 1. *Let (u, c, μ) be a weak solution according to Definition 1. Let assumptions (A1)–(A3) hold, along with (1.11). Then, if*

$$u \in L^{\frac{2}{1+\alpha}+\delta}(0,T;C^{\alpha}(\mathbb{T}^3)), \text{ for } \alpha > 1/3 \text{ and } \delta > 0, \tag{1.13}$$

the following energy equality holds true:

$$E(t) + \int_0^T \int_{\mathbb{T}^3} \left(\nu|\mathbb{D}u|^2 + m|\nabla\mu|^2\right) dx\,dt = E(0), \quad \forall\ t \geq 0.$$

2. Proof of Theorem 1

Let $\rho \in C_0^{\infty}(\mathbb{R}^3)$ be a standard mollifier, i.e. $\rho \geq 0$, with supp $\rho \subset B(0,1) \subset \mathbb{R}^3$, such that $\rho(x) = \rho(|x|)$ and $\int_{\mathbb{R}^3} \rho(x)\,\mathrm{d}x = 1$. For $\epsilon \in (0,1]$, define $\rho_\epsilon(x) := \epsilon^{-3}\rho(\epsilon^{-1}x)$. Then, for any function $f \in L^1_{\mathrm{loc}}(\mathbb{R}^3)$ we define by the convolution

$$f_\epsilon(x) := \int_{\mathbb{R}^3} \rho_\epsilon(x-y)f(y)\,\mathrm{d}y = \int_{\mathbb{R}^3} \rho_\epsilon(y)f(x-y)\,\mathrm{d}y.$$

If $f \in L^1(\mathbb{T}^3)$, then $f \in L^1_{\mathrm{loc}}(\mathbb{R}^3)$, and it turns out that $f_\epsilon \in C^{\infty}(\mathbb{T}^3)$ is 2π–periodic along the x_j–direction, for $j = 1, 2, 3$. Moreover, if f is a divergence–free vector field, then f_ϵ is a smooth divergence–free vector field.

In the following lemma, we collect some properties that will be used throughout this work and for a proof we refer to [5, Section 3].

Lemma 2. *If $u \in C^{\alpha}(\mathbb{T}^3) \cap L^1_{\mathrm{loc}}(\mathbb{T}^3)$, for some $\alpha \in (0,1)$, then*

$$\max_{x\in\mathbb{T}^3} |u(x+y) - u(x)| \leq [u]_\alpha |y|^\alpha, \tag{2.1}$$

$$\max_{x\in\mathbb{T}^3} |u - u_\epsilon| \leq [u]_\alpha \epsilon^\alpha, \tag{2.2}$$

$$\max_{x\in\mathbb{T}^3} |\nabla u_\epsilon(x)| \leq C[u]_\alpha \epsilon^{\alpha-1}. \tag{2.3}$$

In the sequel, we will also use the Constantin–E–Titi commutator from [6]:

$$(u\otimes u)_\epsilon = u_\epsilon \otimes u_\epsilon + r_\epsilon(u,u) - (u-u_\epsilon)\otimes(u-u_\epsilon), \tag{2.4}$$

where

$$r_\epsilon(u,u) := \int \rho_\epsilon(y)[u(x-y)-u(x)]\otimes[u(x-y)-u(x)]\mathrm{d}y.$$

Throughout this work, we set $\gamma = 1$ for convenience, since it does not affect our analysis.

We test (1.1) against $\phi = (u_\epsilon)_\epsilon$. The argument would require a further smoothing with respect to time, which is standard (cf. [4]) but not essential for the proof. Then, we get:

$$\begin{aligned} \frac{1}{2}\|u_\epsilon(T)\|^2_{L^2(\mathbb{T}^3)} - \int_0^T \int_{\mathbb{T}^3} (u\otimes u)_\epsilon : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t + \int_0^T \int_{\mathbb{T}^3} \nu|\nabla u_\epsilon|^2 \mathrm{d}x\mathrm{d}t \\ = \frac{1}{2}\|u_\epsilon(0)\|^2_{L^2(\mathbb{T}^3)} + \int_0^T \int_{\mathbb{T}^3} (\nabla c\otimes\nabla c)_\epsilon : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t \end{aligned} \tag{2.5}$$

and estimate the second term on the left–hand side and the second term on the right–hand side. For the former, we have:

$$\begin{aligned} I_1 &:= \int_0^T \int_{\mathbb{T}^3} (u \otimes u)_\epsilon : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t \\ &= \int_0^T \int_{\mathbb{T}^3} [u_\epsilon \otimes u_\epsilon + r_\epsilon(u,u) - (u-u_\epsilon) \otimes (u-u_\epsilon)] : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t \\ &= \int_0^T \int_{\mathbb{T}^3} r_\epsilon(u,u) : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t - \int_0^T \int_{\mathbb{T}^3} (u-u_\epsilon) \otimes (u-u_\epsilon) : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t =: I_{11} + I_{12}, \end{aligned}$$

since u_ϵ is divergence–free and thus

$$\int_0^T \int_{\mathbb{T}^3} u_\epsilon \otimes u_\epsilon : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t = 0.$$

Regarding the latter:

$$\begin{aligned} I_2 &:= \int_0^T \int_{\mathbb{T}^3} (\nabla c \otimes \nabla c)_\epsilon : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t \\ &= -\int_0^T \int_{\mathbb{T}^3} (\nabla c_\epsilon \otimes \nabla c_\epsilon - (\nabla c \otimes \nabla c)_\epsilon) : \nabla u_\epsilon \mathrm{d}x\mathrm{d}t - \int_0^T \int_{\mathbb{T}^3} u_\epsilon \cdot \mathrm{div}(\nabla c_\epsilon \otimes \nabla c_\epsilon) \mathrm{d}x\mathrm{d}t \\ &=: I_{21} + I_{22}, \end{aligned}$$

where

$$I_{21} = -\int_0^T \int_{\mathbb{T}^3} (\Delta c_\epsilon \nabla c_\epsilon - (\Delta c \nabla c)_\epsilon + \nabla c_\epsilon \cdot \nabla (\nabla c)_\epsilon - \nabla c \cdot \nabla (\nabla c)_\epsilon) \cdot u_\epsilon \mathrm{d}x\mathrm{d}t,$$

while due to (1.3)

$$\begin{aligned} I_{22} &= -\int_0^T \int_{\mathbb{T}^3} (f(c)-\mu)_\epsilon u_\epsilon \cdot \nabla c_\epsilon \mathrm{d}x\mathrm{d}t - \int_0^T \int_{\mathbb{T}^3} u_\epsilon \cdot (\nabla c_\epsilon \cdot \nabla (\nabla c)_\epsilon) \mathrm{d}x\mathrm{d}t \\ &= -\int_0^T \int_{\mathbb{T}^3} (f(c)-\mu)_\epsilon u_\epsilon \cdot \nabla c_\epsilon \mathrm{d}x\mathrm{d}t - \int_0^T \int_{\mathbb{T}^3} u_\epsilon \cdot \nabla \left(\frac{1}{2} |\nabla c_\epsilon|^2 \right) \mathrm{d}x\mathrm{d}t \\ &\overset{(1.2)}{=} -\int_0^T \int_{\mathbb{T}^3} (f(c)-\mu)_\epsilon u_\epsilon \cdot \nabla c_\epsilon \mathrm{d}x\mathrm{d}t. \end{aligned}$$

Therefore,

$$
\begin{aligned}
I_{22} &= -\int_0^T\int_{\mathbb{T}^3} f(c)_\epsilon u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t + \int_0^T\int_{\mathbb{T}^3} \mu_\epsilon u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t\\
&= -\int_0^T\int_{\mathbb{T}^3} (f(c)_\epsilon - f(c_\epsilon))u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t - \int_0^T\int_{\mathbb{T}^3} f(c_\epsilon)u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t\\
&\quad + \int_0^T\int_{\mathbb{T}^3} \mu_\epsilon u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t\\
&= -\int_0^T\int_{\mathbb{T}^3} (f(c)_\epsilon - f(c_\epsilon))u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t - \int_0^T\int_{\mathbb{T}^3} u_\epsilon\cdot\nabla F(c_\epsilon) \mathrm{d}x\mathrm{d}t\\
&\quad + \int_0^T\int_{\mathbb{T}^3} \mu_\epsilon u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t\\
&\overset{(1.2)}{=} -\int_0^T\int_{\mathbb{T}^3} (f(c)_\epsilon - f(c_\epsilon))u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t + \int_0^T\int_{\mathbb{T}^3} \mu_\epsilon u_\epsilon\cdot\nabla c_\epsilon \mathrm{d}x\mathrm{d}t\\
&:= I_{221} + I_{222}.
\end{aligned}
$$

Putting everything together, (2.5) reads:

$$
\frac{1}{2}\|u_\epsilon(T)\|^2_{L^2(\mathbb{T}^3)} + I_{11} + I_{12} + \int_0^T\int_{\mathbb{T}^3} \nu|\nabla u_\epsilon|^2 \mathrm{d}x\mathrm{d}t = \frac{1}{2}\|u_\epsilon(0)\|^2_{L^2(\mathbb{T}^3)} + I_{21} + I_{221} + I_{222}. \tag{2.6}
$$

Now, test (1.3) against $(\mu_\epsilon)_\epsilon$, to obtain:

$$
\int_0^T\int_{\mathbb{T}^3} \mu_\epsilon\left(\partial_t c_\epsilon + \mathrm{div}(cu)_\epsilon - \mathrm{div}(m\nabla\mu)_\epsilon\right)\mathrm{d}x\mathrm{d}t = 0, \tag{2.7}
$$

where

$$
\begin{aligned}
J_1 &:= \int_0^T\int_{\mathbb{T}^3} \mu_\epsilon\partial_t c_\epsilon \mathrm{d}x\mathrm{d}t\\
&\overset{(1.4)}{=} \int_0^T\int_{\mathbb{T}^3} (f(c)_\epsilon - \Delta c_\epsilon)\partial_t c_\epsilon \mathrm{d}x\mathrm{d}t\\
&= \int_0^T\int_{\mathbb{T}^3} (f(c)_\epsilon - f(c_\epsilon))\partial_t c_\epsilon \mathrm{d}x\mathrm{d}t + \int_0^T\int_{\mathbb{T}^3} (f(c_\epsilon) - \Delta c_\epsilon)\partial_t c_\epsilon \mathrm{d}x\mathrm{d}t\\
&= \int_0^T\int_{\mathbb{T}^3} (f(c)_\epsilon - f(c_\epsilon))\partial_t c_\epsilon \mathrm{d}x\mathrm{d}t + \int_0^T\int_{\mathbb{T}^3} \partial_t\left(F(c_\epsilon) + \frac{1}{2}|\nabla c_\epsilon|^2\right)\mathrm{d}x\mathrm{d}t\\
&=: J_{11} + J_{12}.
\end{aligned}
$$

Moreover,

$$\begin{aligned} J_2 &:= \int_0^T \int_{\mathbb{T}^3} \mu_\epsilon \operatorname{div}(cu)_\epsilon \mathrm{d}x\mathrm{d}t \\ &\overset{(1.2)}{=} \int_0^T \int_{\mathbb{T}^3} \nabla\mu_\epsilon \cdot (c_\epsilon u_\epsilon - (cu)_\epsilon)\mathrm{d}x\mathrm{d}t - \int_0^T \int_{\mathbb{T}^3} \mu_\epsilon u_\epsilon \cdot \nabla c_\epsilon \mathrm{d}x\mathrm{d}t \\ &=: J_{21} + J_{22} \end{aligned}$$

and

$$J_3 := -\int_0^T \int_{\mathbb{T}^3} \mu_\epsilon \operatorname{div}(m\nabla\mu)_\epsilon \mathrm{d}x\mathrm{d}t = \int_0^T \int_{\mathbb{T}^3} m|\nabla\mu_\epsilon|^2 \mathrm{d}x\mathrm{d}t$$

and putting everything together, (2.7) reads

$$J_{11} + J_{12} + J_{21} + J_{22} + J_3 = 0. \tag{2.8}$$

Then, we add up (2.6) and (2.8) and let $\epsilon \to 0$.

Notice that I_{222} cancels out with J_{22} and we start with I_{12}:

$$\begin{aligned} I_{12} &\le \int_0^T \int_{\mathbb{T}^3} |u - u_\epsilon|^2 |\nabla u_\epsilon| \mathrm{d}x\mathrm{d}t \\ &= \int_0^T \int_{\mathbb{T}^3} |u - u_\epsilon|^{1+\alpha} |u - u_\epsilon|^{1-\alpha} |\nabla u_\epsilon|^\alpha |\nabla u_\epsilon|^{1-\alpha} \mathrm{d}x\mathrm{d}t, \end{aligned}$$

and making use of (2.1) and (2.2) and after an application of Hölder's inequality

$$\begin{aligned} I_{12} &\le \int_0^T \int_{\mathbb{T}^3} |u - u_\epsilon|^{1+\alpha} \|u(t)\|^{1-\alpha}_{C^\alpha(\mathbb{T}^3)} \epsilon^{\alpha(1-\alpha)} C \|u\|^\alpha_{C^\alpha(\mathbb{T}^3)} \epsilon^{\alpha(\alpha-1)} |\nabla u_\epsilon|^{1-\alpha} \mathrm{d}x\mathrm{d}t \\ &= C \int_0^T \|u\|_{C^\alpha(\mathbb{T}^3)} \left(\int_{\mathbb{T}^3} |u - u_\epsilon|^{1+\alpha} |\nabla u_\epsilon|^{1-\alpha} \mathrm{d}x \right) \mathrm{d}t \\ &\le C \int_0^T \|u\|_{C^\alpha(\mathbb{T}^3)} \|u - u_\epsilon\|^{1+\alpha}_{L^2(\mathbb{T}^3)} \|\nabla u_\epsilon\|^{1-\alpha}_{L^2(\mathbb{T}^3)} \mathrm{d}t. \end{aligned}$$

The fact that u and thus u_ϵ are both in $L^\infty(0,T;L^2(\mathbb{T}^3))$, implies that $\|u - u_\epsilon\|_{L^2(\mathbb{T}^3)} \le C$ for almost every $t \in (0,T)$, for some $C > 0$ independent of ϵ. Thus,

$$\begin{aligned} I_{12} &\le C \int_0^T \|u\|_{C^\alpha(\mathbb{T}^3)} \|\nabla u_\epsilon\|^{1-\alpha}_{L^2(\mathbb{T}^3)} \mathrm{d}t \\ &\le C \|u\|_{L^{\frac{2}{1+\alpha}}(0,T;C^\alpha(\mathbb{T}^3))} \|\nabla u\|^{1-\alpha}_{L^2(0,T;L^2(\mathbb{T}^3))}, \end{aligned}$$

where in the second inequality we have used Hölder's inequality with exponents $\frac{1+\alpha}{2}$ and $\frac{1-\alpha}{2}$. Therefore, the integral is finite and since $(u - u_\epsilon) \otimes (u - u_\epsilon) : \nabla u_\epsilon \to 0$ a.e. by properties of convolutions, we can use the dominated convergence theorem to show that $I_{12} \to 0$. The treatment of I_{11} is identical, but instead of (2.2), we use (2.1).

Since $\nabla c \in L^2(0,T;W^{1,6}(\mathbb{T}^3)) \cap L^\infty(0,T;L^3(\mathbb{T}^3))$, which implies $\Delta c \nabla c \in L^1(0,T;L^2(\mathbb{T}^3))$ and $u \in L^\infty(0,T;L^2(\mathbb{T}^3))$, we directly obtain that $I_{21} \to 0$.

Moreover, the terms I_{221}, J_{11} and J_{21} go to zero as $\epsilon \to 0$. Indeed:

$$I_{221} \leq \|\nabla c\|_{L^\infty(0,T;L^2(\mathbb{T}^3))} \|u\|_{L^2(0,T;L^3(\mathbb{T}^3))} \|f(c)_\epsilon - f(c_\epsilon)\|_{L^2(0,T;L^6(\mathbb{T}^3))} \to 0.$$

Regarding J_{11}, using equations (1.2) and (1.3), we get

$$\begin{aligned} J_{11} &= -\int_0^T \int_{\mathbb{T}^3} (f(c)_\epsilon - f(c_\epsilon))(u \cdot \nabla c)_\epsilon \mathrm{d}x\mathrm{d}t - m \int_0^T \int_{\mathbb{T}^3} \nabla (f(c)_\epsilon - f(c_\epsilon)) \cdot \nabla \mu_\epsilon \mathrm{d}x\mathrm{d}t \\ &=: J_{111} + J_{112}. \end{aligned}$$

Clearly, $J_{111} \to 0$ similar to I_{221} and since $f' \in C([-1,1])$ and $\nabla \mu \in L^2(0,T;L^2(\mathbb{T}^3))$, $J_{112} \to 0$. The fact that $\mu \in L^2(0,T;L^6(\mathbb{T}^3))$, $u \in L^\infty(0,T;L^2(\mathbb{T}^3))$ and $\nabla c \in L^\infty(0,T;L^3(\mathbb{T}^3))$ yield

$$\begin{aligned} J_{21} &\overset{(1.2)}{=} -\int_0^T \int_{\mathbb{T}^3} \mu_\epsilon (u_\epsilon \nabla c_\epsilon - (u \nabla c)_\epsilon) \mathrm{d}x\mathrm{d}t \\ &\leq \|\mu\|_{L^2(0,T;L^6(\mathbb{T}^3))} \|u_\epsilon \nabla c_\epsilon - (u \nabla c)_\epsilon\|_{L^2(0,T;L^{6/5}(\mathbb{T}^3)} \to 0. \end{aligned}$$

Finally, we show that

$$\begin{aligned} J_{12} &= \int_{\mathbb{T}^3} \left(F(c_\epsilon(T)) + \frac{1}{2} |\nabla c_\epsilon(T)|^2 \right) \mathrm{d}x - \int_{\mathbb{T}^3} \left(F(c_\epsilon^0) + \frac{1}{2} |\nabla c_\epsilon^0|^2 \right) \mathrm{d}x \\ &\to \int_{\mathbb{T}^3} \left(F(c(T)) + \frac{1}{2} |\nabla c(T)|^2 \right) \mathrm{d}x - \int_{\mathbb{T}^3} \left(F(c^0) + \frac{1}{2} |\nabla c^0|^2 \right) \mathrm{d}x. \end{aligned}$$

Indeed, for the terms involving F, we have that $c_\epsilon(x,T) \to c(x,T)$ almost everywhere on $\mathbb{T}^3$ and due to the continuity of F, $F(c_\epsilon(x,T)) \to F(c(x,T))$ almost everywhere on $\mathbb{T}^3$. Moreover, $|F(c_\epsilon(x,T))| \leq C$ for some positive constant C independent of ϵ, since c_ϵ is bounded and F is continuous. The dominated convergence theorem can, then, be applied because constants are integrable on the torus. For the gradient terms, notice that due to (1.7), ∇c is in $L^2(\mathbb{T}^3)$ uniformly in t and thus $\nabla c_\epsilon(x,t) \to \nabla c(x,t)$ in $L^2(\mathbb{T}^3)$ for all $t \in [0,T]$ by convolution properties, in particular for $t = T$. Hence,

$$\begin{aligned} &\left| \int_{\mathbb{T}^3} |\nabla c_\epsilon(x,T)|^2 \mathrm{d}x - \int_{\mathbb{T}^3} |\nabla c(x,T)|^2 \mathrm{d}x \right| \\ &\leq \int_{\mathbb{T}^3} |\nabla c_\epsilon(x,T) - \nabla c(x,T)| |\nabla c_\epsilon(x,T) + \nabla c(x,T)| \mathrm{d}x \\ &\leq \|\nabla c_\epsilon(x,T) - \nabla c(x,T)\|_{L^2(\mathbb{T}^3)} \|\nabla c_\epsilon(x,T) + \nabla c(x,T)\|_{L^2(\mathbb{T}^3)} \\ &\leq 2 \|\nabla c(x,T)\|_{L^2(\mathbb{T}^3)} \|\nabla c_\epsilon(x,T) - \nabla c(x,T)\|_{L^2(\mathbb{T}^3)} \to 0. \end{aligned}$$

The kinetic energy terms and the terms at $t = 0$ are treated in a similar way.
Putting everything together, we conclude the proof of the theorem.

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Declarations

Conflict of interest The author has no conflict of interest to report.

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